

STUTTGART ECHTERDINGEN, GermanyWMO: **107380**Lat: **48.6883N**Lon: **9.2236E**Elev: **391**StdP: **96.71**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.2	-13.9	1.2	-8.8	-10.8	1.6	-6.2	10.9	6.5	9.7	6.5	2.1	40	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.1	19.6	29.1	19.0	27.2	18.5	20.8	28.1	20.0	26.9	19.3	25.8	2.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.7	23.3	17.2	12.9	22.3	16.9	12.7	21.9	61.7	27.8	59.2	27.1	56.6	25.6	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.2	6.2	-12.6	34.6	2.8	1.2	-14.6	35.5	-16.2	36.2	-17.7	36.9	-19.7	37.7		
(5)			-13.0	22.4	2.8	0.9	-15.0	23.1	-16.6	23.6	-18.2	24.1	-20.2	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	1.2	1.8	5.8	10.6	14.0	18.1	19.9	18.9	15.3	10.7	5.7	2.6
	DBStd	7.60	4.41	4.39	4.25	3.85	3.67	3.58	3.26	3.31	3.17	3.75	3.74	4.06
	HDD10.0	1094	274	230	138	39	8	1	0	0	1	38	136	230
	HDD18.3	3062	531	463	389	232	142	48	22	33	101	236	378	488
	CDD10.0	1251	1	1	7	57	131	244	306	276	159	61	8	1
	CDD18.3	177	0	0	0	0	7	41	69	51	9	0	0	0
	CDH23.3	1753	0	0	0	18	89	399	673	475	94	5	0	0
	CDH26.7	538	0	0	0	1	17	113	236	155	16	0	0	0
Wind	WSAvg	2.5	2.8	2.7	3.0	2.6	2.7	2.4	2.4	2.1	2.1	2.0	2.3	2.5
Precipitation	PrecAvg	718	43	39	45	53	79	95	76	76	57	49	55	46
	PrecMax	1111	121	122	157	134	167	196	179	200	146	150	125	114
	PrecMin	461	4	8	8	10	27	14	8	17	16	3	9	3
	PrecStd	137	25	22	28	30	36	42	35	40	28	33	27	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.1	19.2	25.2	29.0	32.8	34.1	33.2	28.9	23.9	18.1	13.8
		MCWB	9.5	8.6	10.8	14.3	17.7	20.3	20.3	19.7	18.0	15.9	13.3	10.2
	2%	DB	11.2	11.8	17.1	23.1	26.0	29.9	31.8	30.8	26.0	20.9	15.1	11.8
		MCWB	8.6	7.7	9.6	13.0	16.8	19.4	19.4	19.6	17.7	15.2	11.5	9.2
	5%	DB	9.2	9.8	14.9	21.0	23.2	27.6	29.1	28.0	23.9	19.0	13.2	10.0
		MCWB	7.0	6.8	9.1	12.3	15.0	18.7	19.1	19.3	17.2	14.4	10.5	7.8
	10%	DB	7.2	7.9	12.8	18.2	21.1	25.3	27.1	25.9	21.9	16.9	11.8	8.1
		MCWB	5.7	5.5	8.3	11.2	14.5	17.9	18.6	18.5	16.1	13.4	9.5	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	9.6	11.9	15.0	19.4	21.9	21.7	21.6	19.6	16.8	13.6	10.8
		MCDB	12.7	13.4	17.0	23.4	26.1	29.3	29.7	28.5	25.4	21.5	17.2	12.7
	2%	WB	9.0	8.1	10.5	13.9	17.7	20.4	20.6	20.6	18.5	15.8	11.9	9.5
		MCDB	11.5	11.5	15.0	21.4	24.0	27.2	28.2	27.6	24.2	19.7	14.4	11.5
	5%	WB	7.2	6.8	9.6	12.7	16.4	19.4	19.8	19.8	17.6	14.7	10.7	8.0
		MCDB	8.9	9.2	13.7	19.4	21.5	26.0	27.0	26.6	23.1	17.9	13.2	10.0
	10%	WB	5.7	5.4	8.5	11.7	15.1	18.3	19.1	18.8	16.5	13.8	9.5	6.5
		MCDB	7.1	7.4	12.5	17.7	20.1	24.5	25.7	25.0	20.9	16.6	11.6	7.9
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.6	9.1	10.5	9.8	10.3	10.5	10.3	10.0	9.0	6.6	6.0
		MCDBR	7.7	10.6	14.0	15.7	14.1	14.3	14.8	14.3	13.4	12.7	9.7	8.1
		MCWBR	5.9	7.0	8.0	7.5	6.6	6.3	5.4	5.5	6.5	7.5	6.8	6.1
	5% WB	MCDBR	7.5	9.5	12.0	13.7	12.2	13.0	12.7	13.0	12.1	11.0	9.0	7.7
		MCWBR	5.9	6.7	7.2	6.8	6.0	5.9	5.0	5.4	6.2	7.0	6.6	5.9
Clear-Sky Solar Irradiance	taub	0.304	0.329	0.360	0.394	0.404	0.409	0.409	0.405	0.378	0.353	0.325	0.301	
	taud	2.446	2.387	2.331	2.264	2.266	2.286	2.278	2.309	2.364	2.427	2.460	2.449	
	Ebn at Noon	765	821	850	853	857	854	847	833	820	781	730	718	
	Edh at Noon	64	85	105	124	129	128	127	118	101	81	63	57	
All-Sky Solar Radiation	RadAvg	0.95	1.72	2.90	4.25	5.01	5.71	5.36	4.68	3.46	2.10	1.10	0.76	
	RadStd	0.13	0.28	0.42	0.60	0.54	0.46	0.48	0.45	0.38	0.22	0.14	0.13	